



**Definition of system capacity for center pivots and drip
irrigation at Brazilian coffee regions**

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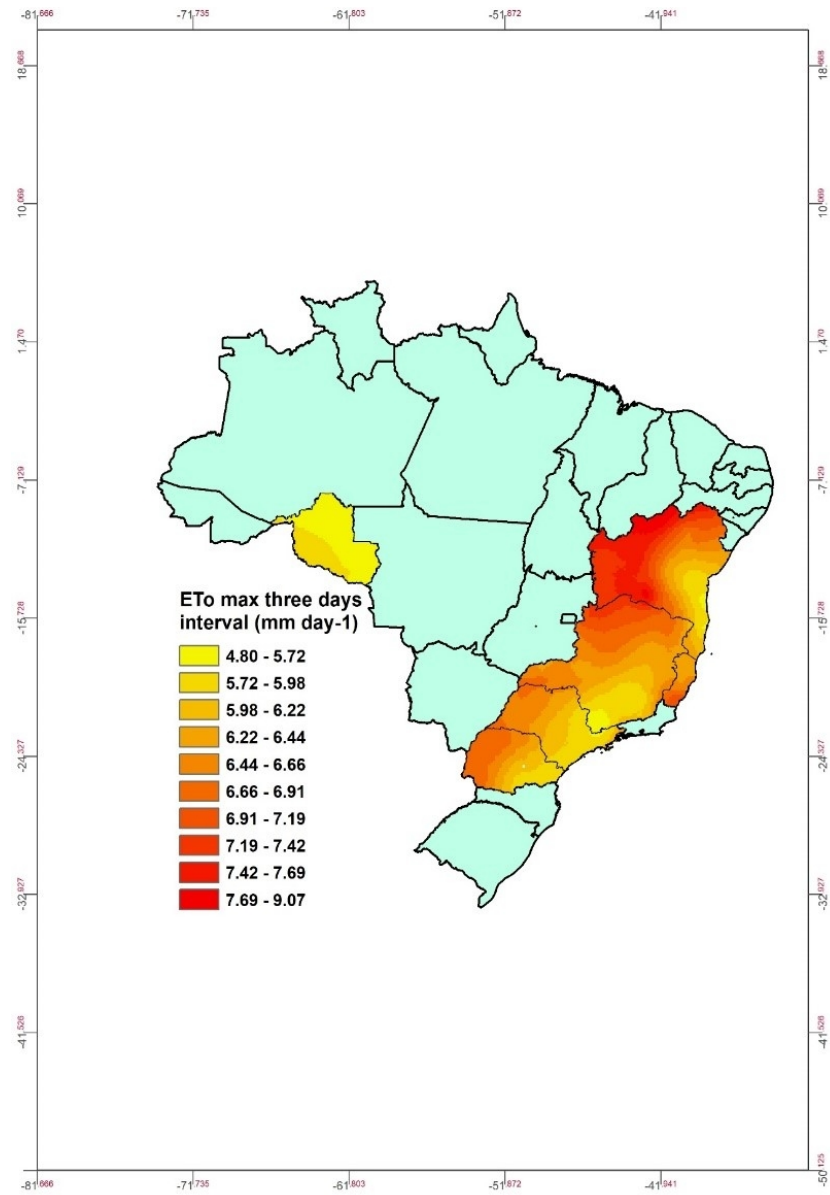


Figure 2: Spatial representation of the maximum daily reference evapotranspiration of the coffee producing states, for 30, 10, 7, 5 and 3 days intervals

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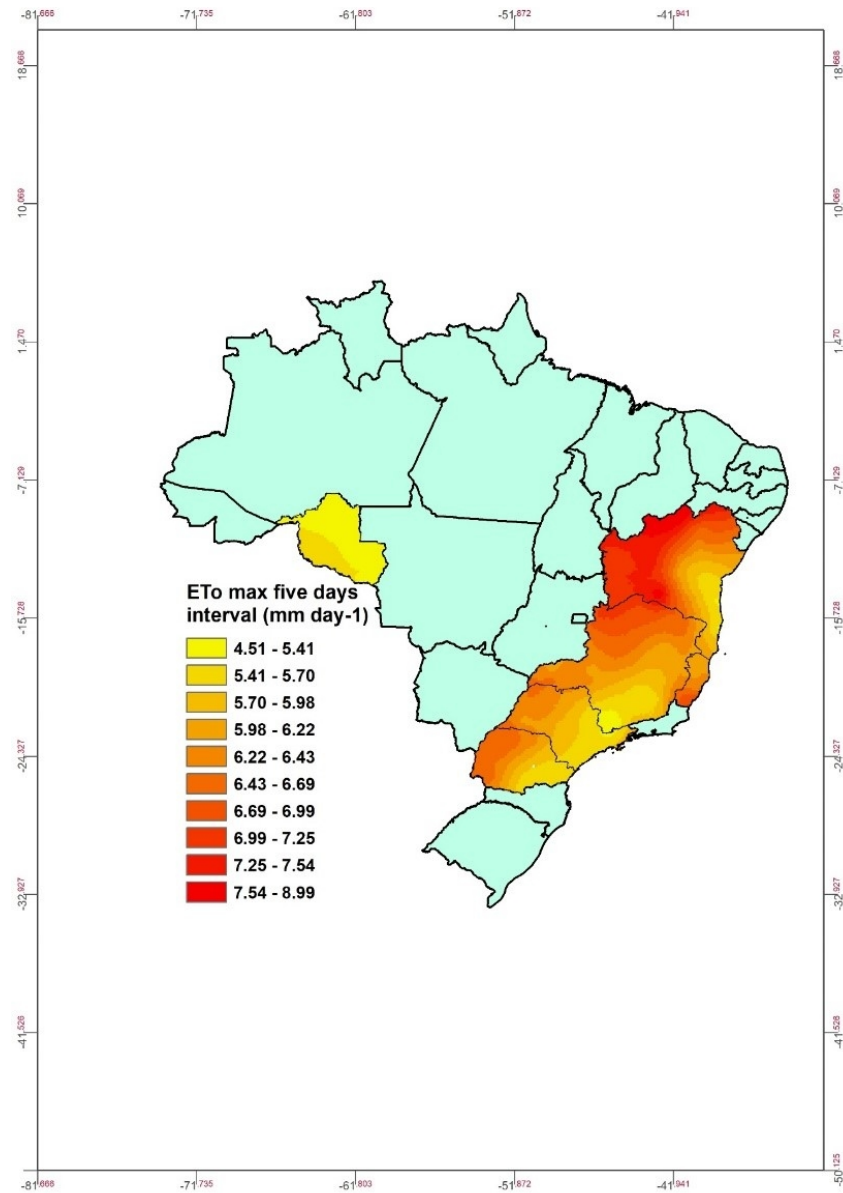


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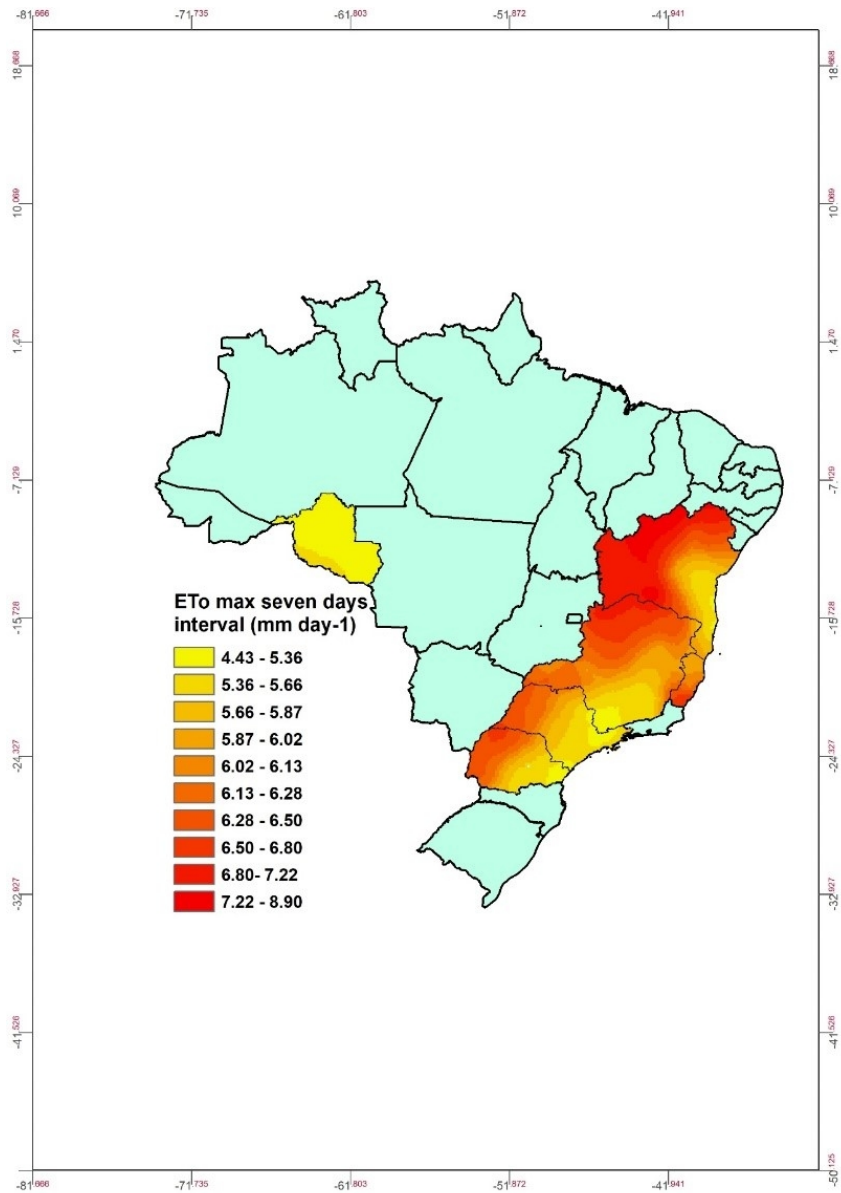


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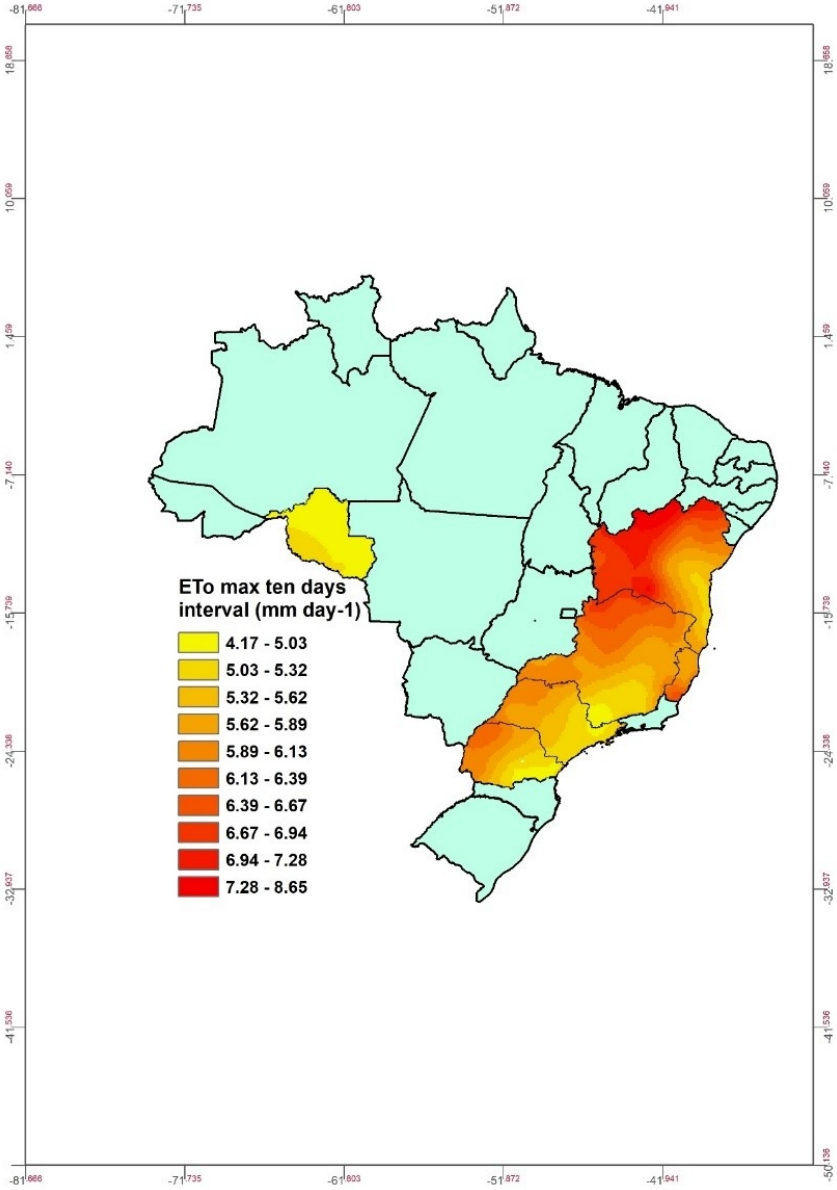


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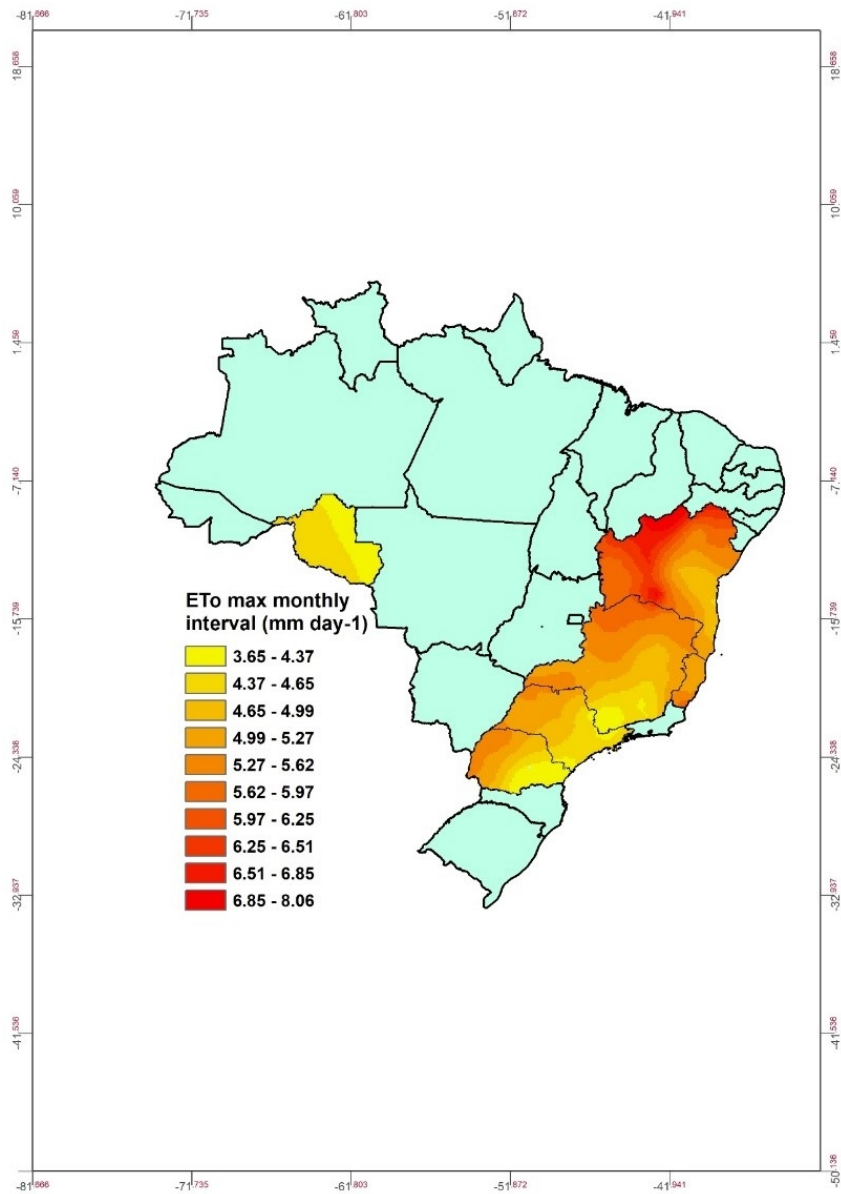


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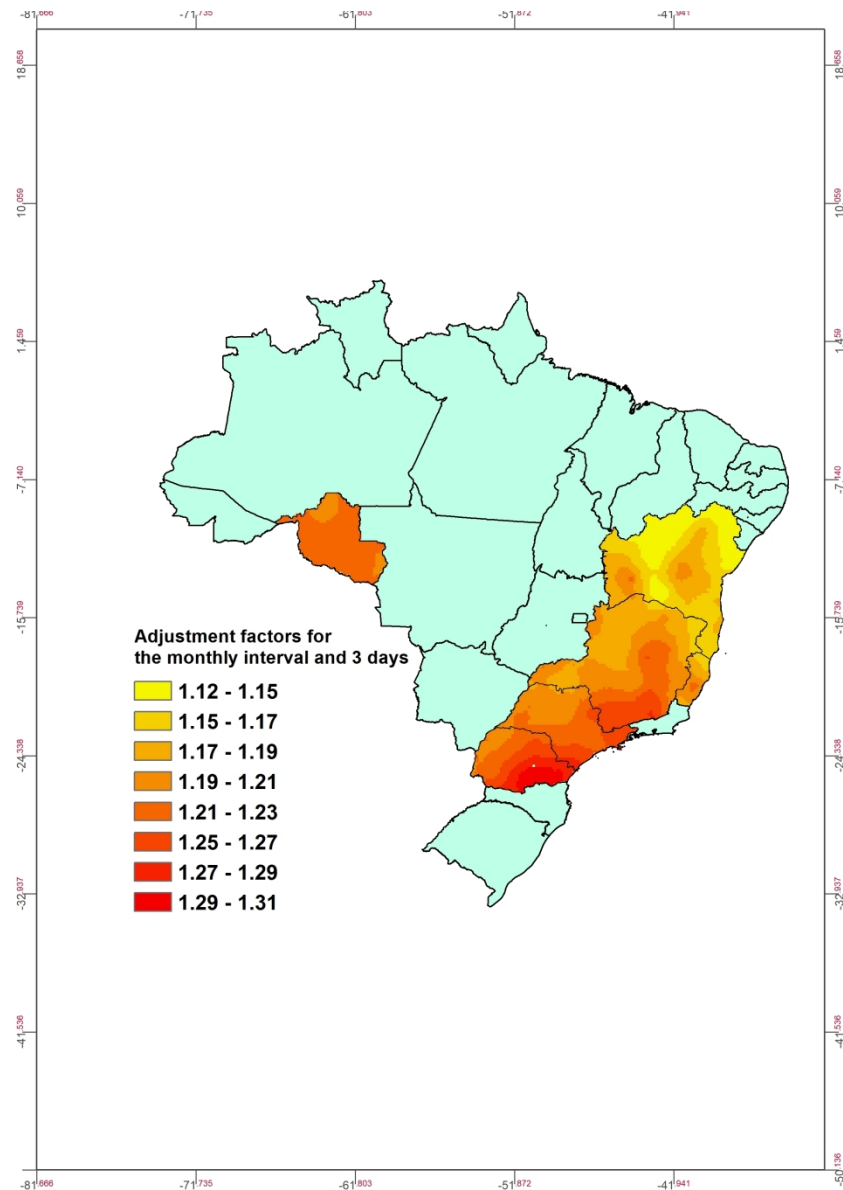


Figure 5: Spatial representation of adjustment factor to correct monthly ET to 3 days moving average.

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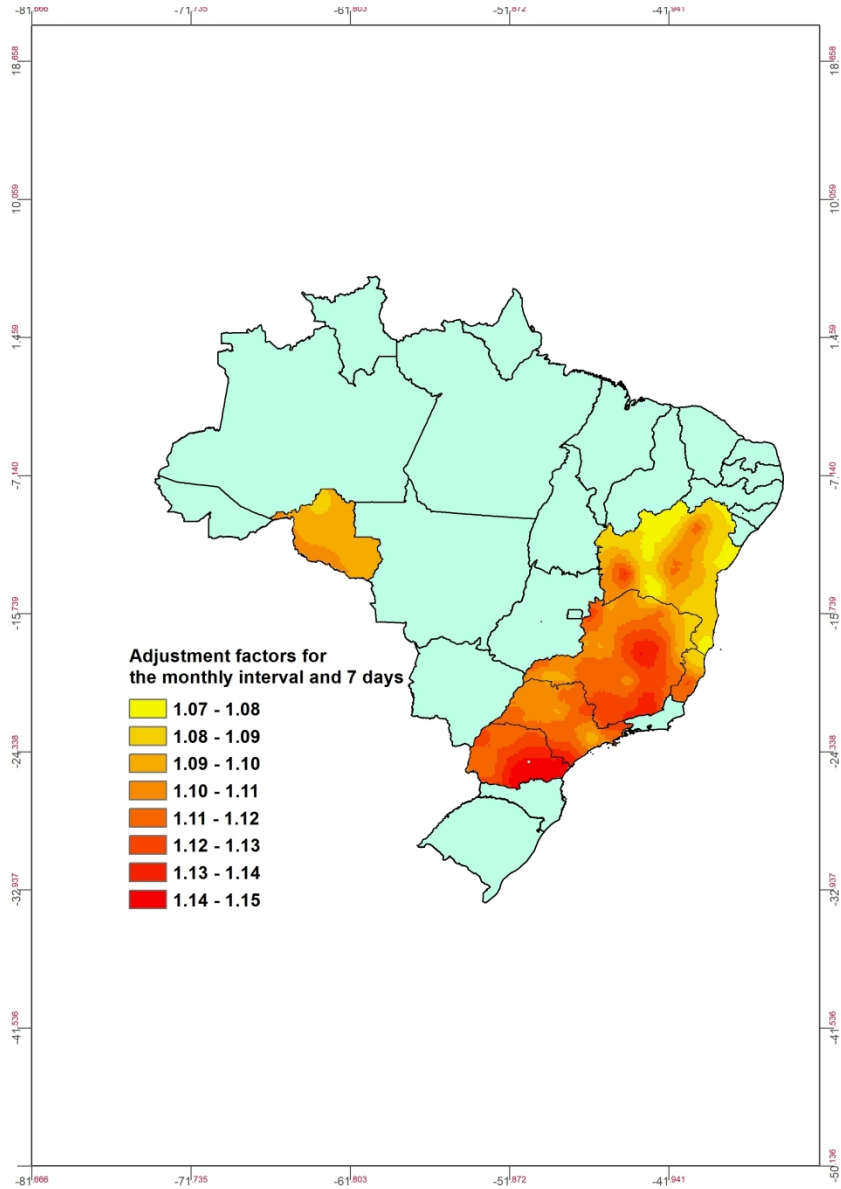


Figure 4: Spatial representation of adjustment factor to correct monthly ET to 7 days moving average.
210x297mm (300 x 300 DPI)

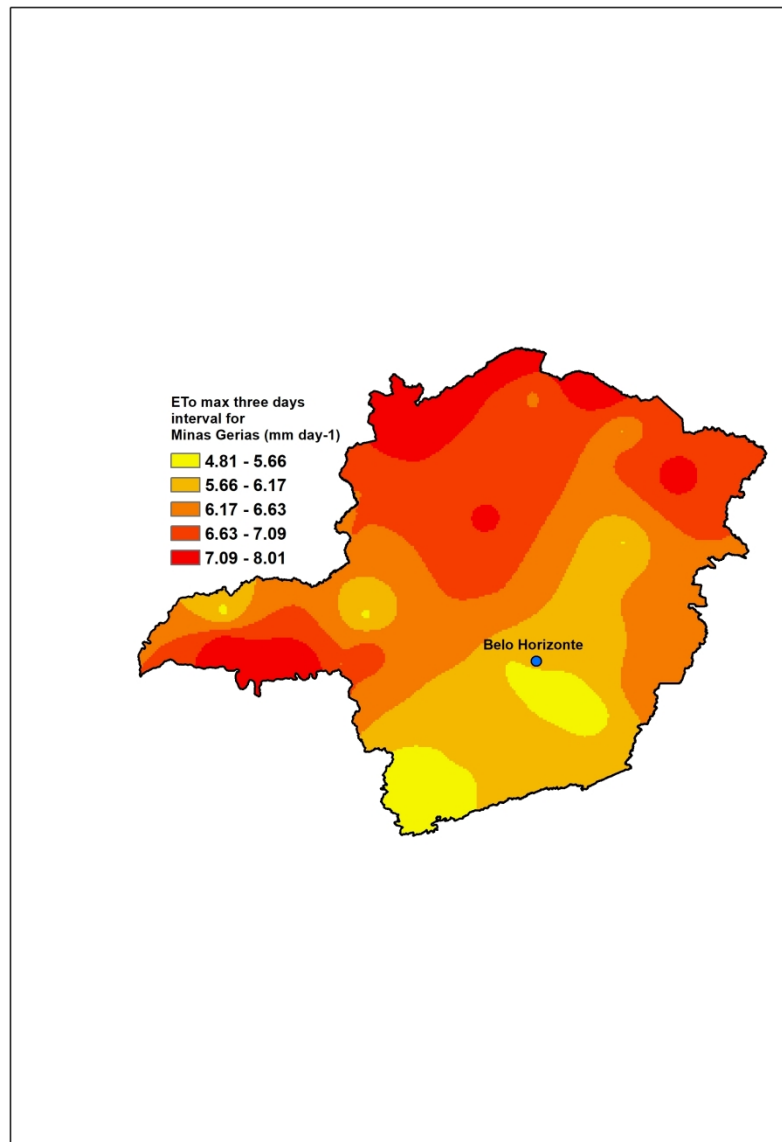


Figure 3: Spatial representation of the maximum daily reference evapotranspiration of the state of Minas Gerais, for the monthly interval and three days

210x297mm (300 x 300 DPI)

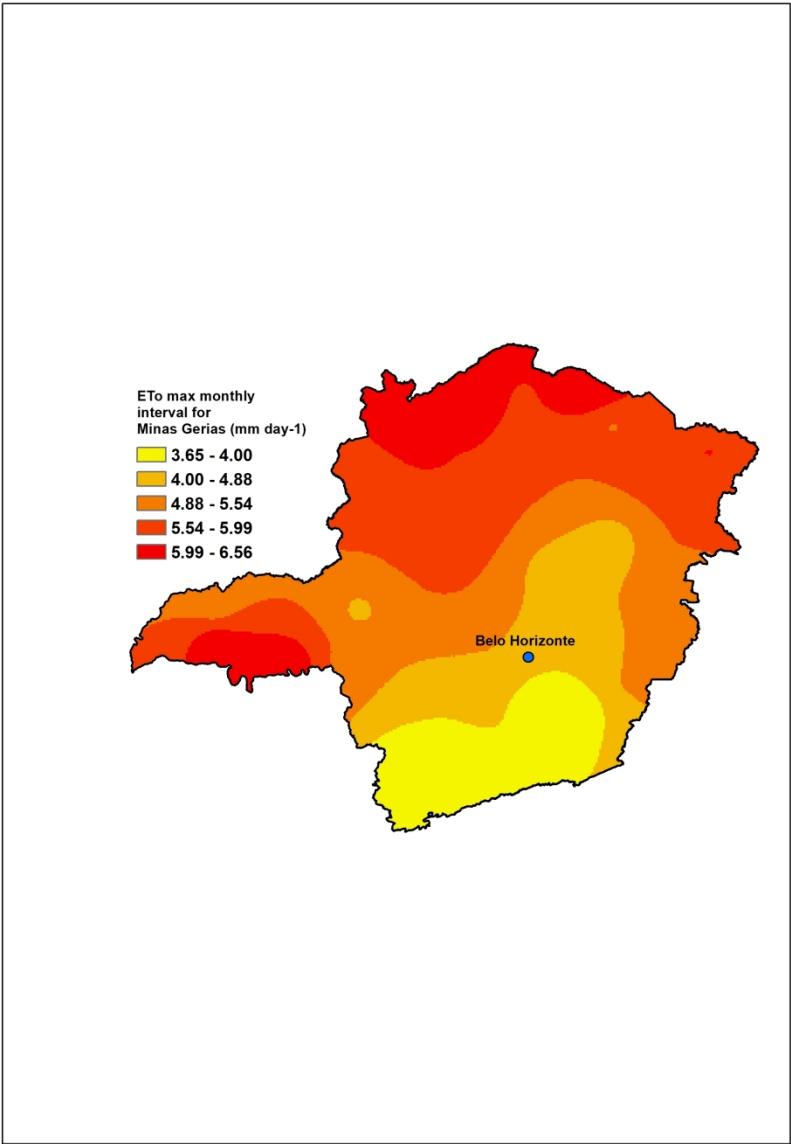


Figure 3: Spatial representation of the maximum daily reference evapotranspiration of the state of Minas Gerais, for the monthly interval and three days

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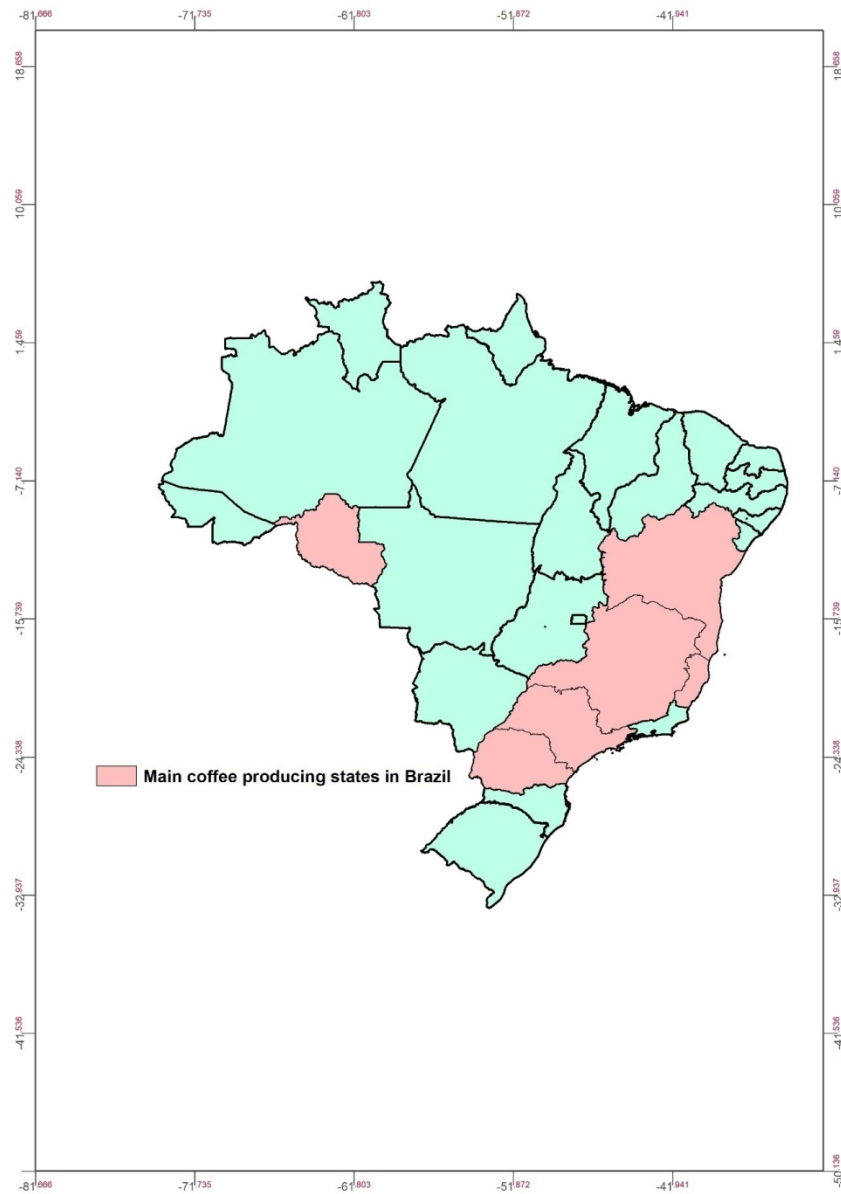


Figure 1: Main Brazilian coffee producing states

150x212mm (220 x 220 DPI)

Definition of system capacity for center pivots and drip irrigation at Brazilian coffee regions

Definição da capacidade de irrigação por sistemas de pivô central e gotejamento para as principais regiões cafeeiras do Brasil

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Abstract

Coffee cultivation stands out among a wide range of agricultural commodities that drive the Brazilian economy. Knowledge of its characteristics and needs is essential to ensure yield and, consequently, sustainability and revenues. Therefore, irrigation plays an important role in coffee production, where the appropriate design of well-designed systems achieve greater efficiency, and consequently provide superior results. This study was designed to estimate evapotranspiration during time intervals of 3, 5, 7 and 10 days at main coffee-producing regions at Brazilian territory. To calculate evapotranspiration, data from 141 meteorological stations distributed throughout Brazilian coffee-producing states were obtained from the National Institute of Meteorology from January 2011 to December 2021. Subsequently, the Penman-Monteith equation (FAO-56) was applied to calculate evapotranspiration. Using Kriging techniques, evapotranspiration maps were generated for coffee-growing regions in Brazil. It was observed that designing irrigation systems based on monthly ET divided by the number of days underestimates the critical evapotranspiration values, which may imply yield losses. The minimum evapotranspiration values were 3.65; 4.17; 4.43; 4.51 and 4.80 mm day⁻¹ for time intervals of 30, 10, 7, 5 and 3 days respectively, and the highest evapotranspiration values are 8.06; 8.65; 8.88; 8.99 and 9.07 mm day⁻¹ for time intervals of 30, 10, 7, 5 and 3 days, respectively. The adjustment factor values for obtaining the 3-day ET period range from 1.12 to 1.31. In conclusion, values obtained for time intervals of 30 days are useful for estimating monthly water use but not for designing system capacity.

Key-words: coffee, irrigation, evapotranspiration

Resumo

A cafeicultura se destaca entre uma ampla gama de commodities agrícolas que movimentam a economia brasileira. O conhecimento de suas características e necessidades é essencial para garantir a produtividade e, consequentemente, a sustentabilidade e as receitas. Portanto, a irrigação desempenha um papel importante na produção de café, onde o projeto adequado de sistemas alcança maior eficiência e, proporciona resultados superiores. Este estudo foi concebido com o propósito de estimar a evapotranspiração durante intervalos de tempo de 3, 5, 7 e 10 dias nas principais regiões produtoras de café no território brasileiro. Para calcular a evapotranspiração, dados de 141 estações meteorológicas distribuídas pelos estados produtores de café do Brasil foram obtidos do Instituto Nacional de Meteorologia de janeiro de 2011 a dezembro de 2021. Posteriormente, a equação de Penman-Monteith (FAO-56) foi aplicada para calcular a evapotranspiração. Utilizando técnicas de Krigagem, mapas de evapotranspiração foram gerados para regiões produtoras de café no Brasil. Observou-se que projetar sistemas de irrigação com base na ET mensal dividida pelo número de dias subestima os valores críticos de evapotranspiração, o que pode implicar perdas de rendimento. Os valores mínimos de evapotranspiração foram 3,65; 4,17; 4,43; 4,51 e 4,80 mm dia⁻¹ para intervalos de tempo de 30, 10, 7, 5 e 3 dias, respectivamente, e os maiores valores de evapotranspiração são 8,06; 8,65; 8,88; 8,99 e 9,07 mm dia⁻¹ para intervalos de tempo de 30, 10, 7, 5 e 3 dias, respectivamente. Os valores do fator de ajuste para obtenção do período de 3 dias da ET variam de 1,12 a 1,31. Em conclusão, os valores obtidos para intervalos de tempo de 30 dias são úteis para estimar o uso mensal de água, mas não para projetar a capacidade do sistema.

Palavras-chave: Café, Irrigação, Evapotranspiração

Introduction

Irrigation has a direct impact on Brazilian economy, and its use is of utmost importance for domestic supply. According to Guimarães (2020), irrigated agriculture is not limited to the domestic market, but is also responsible for a large part of Brazil's exports, contributing positively to the country's trade balance.

The design of an irrigation system begins by determining the crop's water requirement, which can be affected by the climate, soil, and plant's own characteristics, among other factors (BEZERRA, 2024). In fact, Dorenboos and Pruitt (1977) point out that sufficiently long (10 years or more) climatic records must be considered through frequency analysis and the authors proposed a method to adjust the maximum monthly crop water requirement using mean data for four different climatic conditions. One of them, which results in the largest adjustment factor is described as mid-continental climate and sub-humid to humid climates with highly variable cloudiness in month of peak ET. The factors ranged from 1,07 for high water holding capacity (WHC) soils to 1.37 (low WHC soils).

In practice, the irrigation system is designed for the worst situations faced by the crop during its growth and development cycle, mainly to ensure that there is no water deficit (Albuquerque, 2000). According to Alves (2020), water deficit limits plant growth and development, negatively impacting yield, considering that water acts in several physiological processes.

Irrigated agriculture in Brazil is responsible for a large part of production of agricultural commodities, including coffee. According to updated data from the National Supply Company (CONAB), for the year 2024, the Brazilian territory has approximately 1.87 million hectares occupied by coffee, with emphasis on the states of Minas Gerais, Bahia, Espírito Santo, Paraná, São Paulo and Rondônia. However, only a portion of this area is irrigated.

In order to design system capacity, evapotranspiration data must be calculated. The Penman-Monteith equation is adopted as a standard method by the Food and Agriculture Organization of the United Nations (FAO). However, in Brazil, one of the major challenges encountered for designing an irrigation system is related to acquisition and reliability of meteorological data used in the equations. With reliable reference evapotranspiration (ET_o) values, it is possible to determine the gross depth of the system (d) according to equation 1:

$$d = \frac{Kc.Kaj.ET_o}{Ea}$$

(1)

where d is gross depth, K_c is the crop coefficient with maximum values from 1.10 to 1.20 (Doorenbos; Pruitt, 1977; Allen et al., 1998), E_a is the application efficiency of the system (0.85 for center pivot and 0.95 for drip according to ANA, 2021) and K_{aj} is the adjustment factor which in turn is the result of the multiplication of a location factor (K_l) and even a K_s factor influenced by the soil water regime related to phytotechnical aspects, salinity, etc. Under normal conditions K_s = 1.0 and K_{aj} is summarized in the value of K_l that can be obtained, according to Kelleer and Bliesner (1990) as:

$$Kl = \sqrt{Fc}$$

(2)

Where F_c is the fraction of the area shaded by the plant. In systems that wet 100% of the soil, K_l = 1.0. For a center pivot with coffee trees planted in concentric circles and irrigated by LEPA (low energy precision application) nozzles operating with the Bubble 1 or 2 options, the width of the wetted strip varies according to the emitter flow rate. In fact, Ribeiro et al.(data?) found strips width varying from 0.57 to 3.24 meters for LEPA QuadSpray nozzles ranging from sizes 4 to 25. In this case, it is recommended to adopt K_l according to equation 2.

Despite precise definitions at all equations involved, the main challenge relies on precise evapotranspiration values for designing the irrigation system capacity. In analogy to the method proposed by Dorenboos and Pruitt (1977), the present study estimates the maximum reference evapotranspiration for time intervals of 30, 10, 7, 5 and 3 days using the Penman-Monteith equation in the main coffee-growing regions of Brazil. Selection of 3 days interval would be appropriate for low WHC soils while 7 days would be more appropriate for low WHC soils.

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Methodology

Data acquisition

The data for this study were obtained from the Meteorological Database (BDMEP) of the National Institute of Meteorology (INMET). Hourly data from 141 automatic stations (after data processing) distributed in the states of Minas Gerais, Bahia, Espírito Santo, Paraná, São Paulo and Rondônia were used (Figure 1), for the period from January 1, 2011 to December 31, 2021. Automatic stations were chosen due to greater reliability of the measured data, since conventional stations often do not have updated maintenance, which can generate inaccurate measurements.

Data processing

The database of the National Institute of Meteorology subdivides the automatic stations according to their operating status, i.e., it has operating stations, stations that have failed, stations that are closed, stations that are deactivated, and stations that are temporarily closed. Given this fact, the first step was to discard all non-operating stations.

The database also has incomplete stations, i.e., newer stations that were installed after 2011, meaning there is no data for the years proposed by the study. Given this, the second step was to discard these stations.

Automatic stations are instruments that are subject to reading and measurement errors, which can result in inconsistent or missing values. The discards were performed so that errors such as those mentioned above could be eliminated.

To ensure data homogeneity, a consistency analysis was performed using the double accumulated mass model described by Bertoni and Tucci (2007). This methodology consists of selecting the station of interest together with the closest stations. Using the annual totals, a

graph is plotted with the averages of the other stations; there will be consistency when there is a linear trend in the data.

Evapotranspiration Calculation

With the processed data, the daily reference evapotranspiration (ET_o) was calculated using the Penman-Monteith-FAO equation:

$$ET_o = \frac{0,408\Delta(Rn - G) + \gamma \frac{900}{T + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0,34u_2)} \quad (3)$$

where,

ET_o is the reference evapotranspiration (mm d⁻¹);

R_n is the net radiation or radiation balance (MJ m⁻² d⁻¹);

G is the heat flux in the soil, assumed to be zero (for daily calculations it is ignored due to its small magnitude);

T is the average daily air temperature, measured at 2 m height (°C);

u₂ is the wind speed, measured at 2 m height (m s⁻¹);

Δ is the slope of the vapor pressure curve (kPa °C⁻¹);

e_s is the vapor saturation pressure (kPa);

e_a is the real vapor pressure (kPa);

γ is the psychrometric constant (kPa °C⁻¹).

To calculate the independent terms of the Penman-Monteith equation, we have:

$$\gamma = 0,665 \times 10^{-3} P_{atm}$$

(4)

where,

P_{atm} is the atmospheric pressure, kPa.

$$P_{atm} = 101,3 \left(\frac{293 - 0,0065z}{293} \right)^{5,26}$$

(5)

where,

z is the height of the station above sea level, m.

$$\Delta = \frac{4096 \left[0,6108 \exp \frac{17,27T}{T+237,3} \right]}{(T+237,3)^2}$$

(6)

where,

$\exp$ is the natural logarithmic base, 2.7183;

T is the mean air temperature, °C.

$$e^o(T) = 0,6108 \exp \left[\frac{17,27T}{T + 237,3} \right]$$

(7)

where,

$e^o(T)$ is the saturation vapor pressure at a given air temperature, kPa;

T is the air temperature, °C.

$$e_s = \frac{e^o(Tmáx) + e^o(Tmin)}{2} \quad (8)$$

where,

e_s is the vapor saturation pressure, kPa.

$$R_n = R_{ns} - R_{nl} \quad (9)$$

where.

R_{ns} is the net shortwave radiation reaching the earth's surface, MJm²day⁻¹

R_{nl} is the net longwave radiation reaching the earth's surface, MJm²day⁻¹

$$R_{ns} = R_s(1 - \alpha) \quad (10)$$

where,

α is the albedo, 0.23

$$R_{nl} = \alpha \left[\frac{TmáxK^4 + TmínK^4}{2} \right] (0,34 - 0,14\sqrt{e_a}) \left(1,35 \frac{R_s}{R_{so}} - 0,35 \right) \quad (11)$$

where,

σ is the Stefan-Boltzmann constant, 4.903 x 10⁻⁹ MJm² day⁻¹;

T_{maxK} is the absolute maximum temperature over a 24-h period, °K;

T_{minK} is the absolute minimum temperature over a 24-h period, °K;

R_s is the measured or calculated solar radiation, MJ m⁻² day⁻¹;

R_{so} is the measured or calculated clear-sky radiation, MJm⁻² day⁻¹

$$R_{so} = (0,75 + 2 \times 10^{-5} z) R_a$$

(12)

where,

z is the altitude of the station above sea level, m;

Ra is the extraterrestrial radiation, MJm-2 day-1

$$e_a = e^o(T_{min}) = 0,611 \exp \left[\frac{17,27 T_{min}}{T_{min} + 237,3} \right]$$

(13)

where,

ea is the actual vapor pressure, kPa;

e° (Tmin) is the saturation vapor pressure at the minimum temperature, kPa;

Tmin - minimum temperature, °C;

$$R_s = k_{Rs} \sqrt{(T_{max} - T_{min})} R_a$$

(14)

where,

Rs is the solar radiation, MJ m-2 day-1;

Ra is the extraterrestrial radiation, MJ m-2 day-1;

Tmax is the maximum temperature, °C;

Tmin is the minimum temperature, °C;

kRs is the adjustment coefficient, °C-0.5.

Exploratory data analysis

After calculating evapotranspiration and before calculating moving averages and using geostatistics, the data were subjected to exploratory analysis. The main objective at this stage

is to identify discrepant values that could compromise the series and alter the averages, underestimating or overestimating the expected result.

To this end, a graph was generated using ArcGIS[®] software in the Boxplot model, which makes it possible to visualize and identify possible outliers. Using the graph, the most discrepant values were identified to analyze whether or not the data should be excluded. The outlier values were analyzed, but they did not present irregularities, and they are regions where similar values have already been observed in practice. Given this scenario, these data were not excluded and were included in the geostatistics.

Calculation of moving averages

Based on the estimated daily evapotranspiration values, using Microsoft Excel spreadsheets, the moving averages were calculated for 30, 10, 7, 5 and 3 days. The spreadsheet runs through each daily evapotranspiration value, calculating the moving average for the defined interval, i.e., if the critical interval is 3 days, for example, the spreadsheet selects the interval from the 1st to the 3rd day, calculating the simple average of the values of these 3 days (adding and dividing by the number of values), then from day 2 to day 4, and so on, repeating the process for the entire series, which also occurs for the other proposed values, respecting their particularities. With the results obtained, the spreadsheet selects the most critical value of the established time series (January 1, 2011 to December 31, 2021), and then divides it by the duration of time interval, obtaining the daily average.

Map generation

The maps were generated using the ArcGIS[®] tool to estimate evapotranspiration in regions without stations. The simple Kriging interpolation method was used, since it was the best methodology for the proposed work, finding interpolated values for the entire Brazilian

territory, while other models such as ordinary Kriging and IDW did not fit the polygon well, resulting in “empty” spaces.

The chosen model was the Exponential model, which showed satisfactory results regarding the Degree of Spatial Dependence (behavior of the data and its correlation). This index was presented by Cambardella et al. (1994), and is divided into weak, moderate and strong and obtained by the following equation:

$$DE = (\frac{C0}{C1+C0}).100$$

(15)

where,

C1 is the contribution;

C0 is the nugget effect of the model;

DE is the spatial dependence.

Values above 75% are classified as weak, values between 25 and 75% are considered moderate and below 25% are considered strong values for the spherical model. In addition to the calculation of spatial dependence, cross-validation was performed to choose the best model. Table 1 shows the values of parameters for calculating the Degree of Spatial Dependence (GD), nugget effect (C0), contribution (C1), and the regression analysis taking into account the Root Mean Square Error (RMSE), generated in the software itself.

Adjustment factor mapping

An interesting tool in the ArcGis application is the Raster Calculator, which allows users to perform “map operations”. With this in mind, a division operation was performed on the maps above, with the aim of generating correction factors for the monthly average, so that values

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3 closer to the actual water needs of crops can be found. In view of this, the values of the 7 days
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5 and 3 days critical interval were divided by the critical values obtained by the monthly
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7 average. This operation was chosen because, for dimensioning irrigation projects, the use of
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9 weekly evapotranspiration values is already sufficient to estimate the water demand of the
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11 crop.
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Results and discussion

Figure 3 shows the spatial distribution of reference evapotranspiration in the main Brazilian coffee-producing states. The image shows the reference evapotranspiration maps in mm day⁻¹. The location with the lowest evapotranspiration is Monte Verde – MG, with values of 3.65; 4.17; 4.43; 4.51 and 4.80 mm day⁻¹ for 30, 10, 7, 5 and 3 days intervals, respectively. On the other hand, the highest reference evapotranspiration is in the municipality of Remanso – BA, with values of 8.06; 8.65; 8.88; 8.99 and 9.07 mm day⁻¹ for 30, 10, 7, 5 and 3 days intervals, respectively.

As can be seen in the figures, there is an increase on evapotranspiration values as one approaches the northeast region of the country (north of Minas Gerais and state of Bahia). These are locations with higher temperatures when compared to other locations, mainly due to their low relative humidity and high solar radiation rates, as Monteiro (2020) shows in his work.

As an example, this study obtained a value of 5.80 mm day⁻¹ for the monthly average, and 7.21 mm day⁻¹ when using the critical interval of 3 days for the municipality of Barreiras - BA, higher than the results found by Vicente et al. (2015), who, when analyzing the effect of different irrigation depths on a coffee plant irrigated by a center pivot for the same location, found an average reference evapotranspiration value of 5.62 mm day⁻¹ for the period from November 2004 to May 2007.

In the Northeast region, for the municipality of Ilhéus in Bahia, an average value of 4.48 mm day⁻¹ was found based on the 7 days moving average and 4.08 mm day⁻¹ for evapotranspiration based on the monthly average. This result, although close, differs from the values obtained by Da Silva Braga et al. (2023), who found approximate evapotranspiration

values of 4.19 mm day⁻¹ and 3.87 mm day⁻¹ for the years 2017 and 2018 respectively for this same location.

Santos, Souza and Rosa (2020), in their work, found an average evapotranspiration value for the municipality of Maringá in the state of Paraná of 5.22 mm day⁻¹ for the years 2016 to 2018. This value is close to the value found by the present study when using the monthly average, which is 5.64 mm day⁻¹; however, when using 3 days intervals, this value increases to 7.16 mm day⁻¹. The underestimation of the real daily water requirement is also notable.

Nandorf et al. (2020), in their work studying the reference evapotranspiration, found a value of 5.4 mm day⁻¹ for the western Bahia municipality of Luis Eduardo Magalhães. This value is close to the value found for this work when using the monthly average, which was 5.6 mm day⁻¹. Although the values are close and using the same data from INMET, the first author used data for the period between 2004 and 2008. However, when compared to critical interval of 5 days, this value rises to 6.89 mm day⁻¹, making it a clear underestimate when using data based on the monthly average.

The state of Minas Gerais (the largest coffee producing state in Brazil according to CONAB, 2024) presents a variation in reference evapotranspiration of 3.65 to 6.59 mm day⁻¹ in relation to the monthly average; however these values increase when the critical interval decreases, reaching 4.80 and 8.03 mm day⁻¹ for a 3-day average (Figure 3). These values differ from Cunha et al. (2021), who found a range of 2.91 to 4.31 mm day⁻¹ throughout the state. The difference can be explained mainly by the years used for the study, since the second author used data between 2007-2018, however the results corroborate the aspect that evapotranspiration tends to increase from the South to the North of Minas, going against what was exposed by Kool (2014) in his work, on the variables that influence evapotranspiration such as: solar radiation, temperature, vapor pressure deficiency, wind speed, causing the

reference evapotranspiration values to differ with the change in region. For Uberaba region, an average value of 5.7 mm day^{-1} was found based on the monthly average, however, when using the moving average for the 10-day period, a value of 6.5 mm day^{-1} was observed. For the same location, Rosa et al. (2023) found an average value of 4.89 mm day^{-1} , a divergence that can be explained by the difference in the years used for the study since the authors used the period from 2017 to 2020. Also for the state of Minas Gerais, Pinheiro et al. (2019) found an average value of 5.14 mm day^{-1} for the municipality of Mocambinho, while in this study a value of 5.48 mm day^{-1} was found for the monthly average and 6.07 mm day^{-1} for the critical interval of 10 days. Although these are close values (comparing the monthly averages), the first author used the methodology of Hargreaves and Samani (1985), which, although it requires fewer input variables, when well calibrated, presents good results when compared to the Penman-Monteith equation.

Something of great relevance to be considered when making comparisons to other studies presented at literature is the specific study period of each one, as well as the database used. Since these are measurements taken in different hydrological years and with different instrumentation, there is the possibility of divergence. An example of a change would be due to very short series that end up excluding possible climate events (mainly extreme events) whose variables influence the estimate of reference evapotranspiration (precipitation, dry periods, etc.). Thus, although the values are similar and do not present great discrepancies, they do have differences.

In analogy to Dorenboos and Pruitt (1977), adjustment factors were calculated to estimate, from monthly values, the ET for 7 and 3 days long intervals, considering appropriate do high and low WHC soils, respectively. Those values are presented in figures 4 and 5. It can be concluded that values ranged from 1.07 to 1.15 for 7 days period and from 1.12 to 1.31 for 3

days periods, respectively. In fact, these values are quite similar to those proposed by the authors mentioned.

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Conclusion

Considering that monthly evapotranspiration provided by different database are mean values obtained from historical data in a changing climate scenario and that irrigation intervals are typically much less than a month or even a week, this research investigated more appropriate values to design irrigation systems for crop production. The values establish an adjustment factor to obtain the 3-day period value for ET of 1.31 and 1.12, respectively for the main coffee regions of Brazil.

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Author Contribution

Conceptual idea: Luiz Antônio Lima

Methodology design: Luiz Antônio Lima; Adriano Valentim Diotto, José Alves Junqueira Junior; Lincoln Oliveira Mendes

Data collection: Lincoln Oliveira Mendes

Data analysis and interpretation: Lincoln Oliveira Mendes

Writing and editing: Lincoln Oliveira Mendes; Luiz Antônio Lima; Adriano Valentim Diotto, José Alves Junqueira Junior

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Figure 1: Main Brazilian coffee producing states

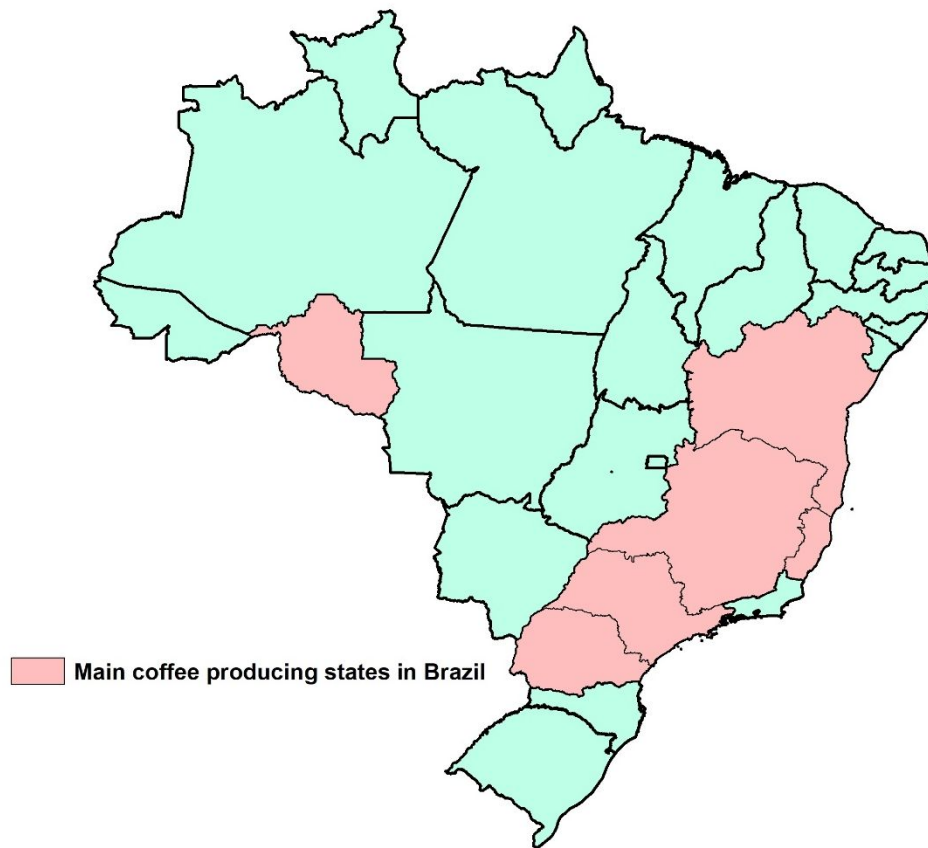


Table 1: Geostatistical model adjustment parameters, classification and regression analysis of cross-validation error

Model	C_1	C_0	GD	Classification	RMSE
Espheric	0.51	0.52	0.504854	Moderate	0.5151
Gaussian	0.45	0.59	0.567308	Moderate	0.5305
Exponential	0.73	0.3	0.291262	Moderate	0.4908

Degree of Spatial Dependence (GD), nugget effect (C_0), contribution (C_1), and the regression analysis taking into account the Root Mean Square Error (RMSE), generated in the software itself.

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Figure 2: Spatial representation of the maximum daily reference evapotranspiration of the coffee producing states, for 30, 10, 7, 5 and 3 days intervals

For Review Only

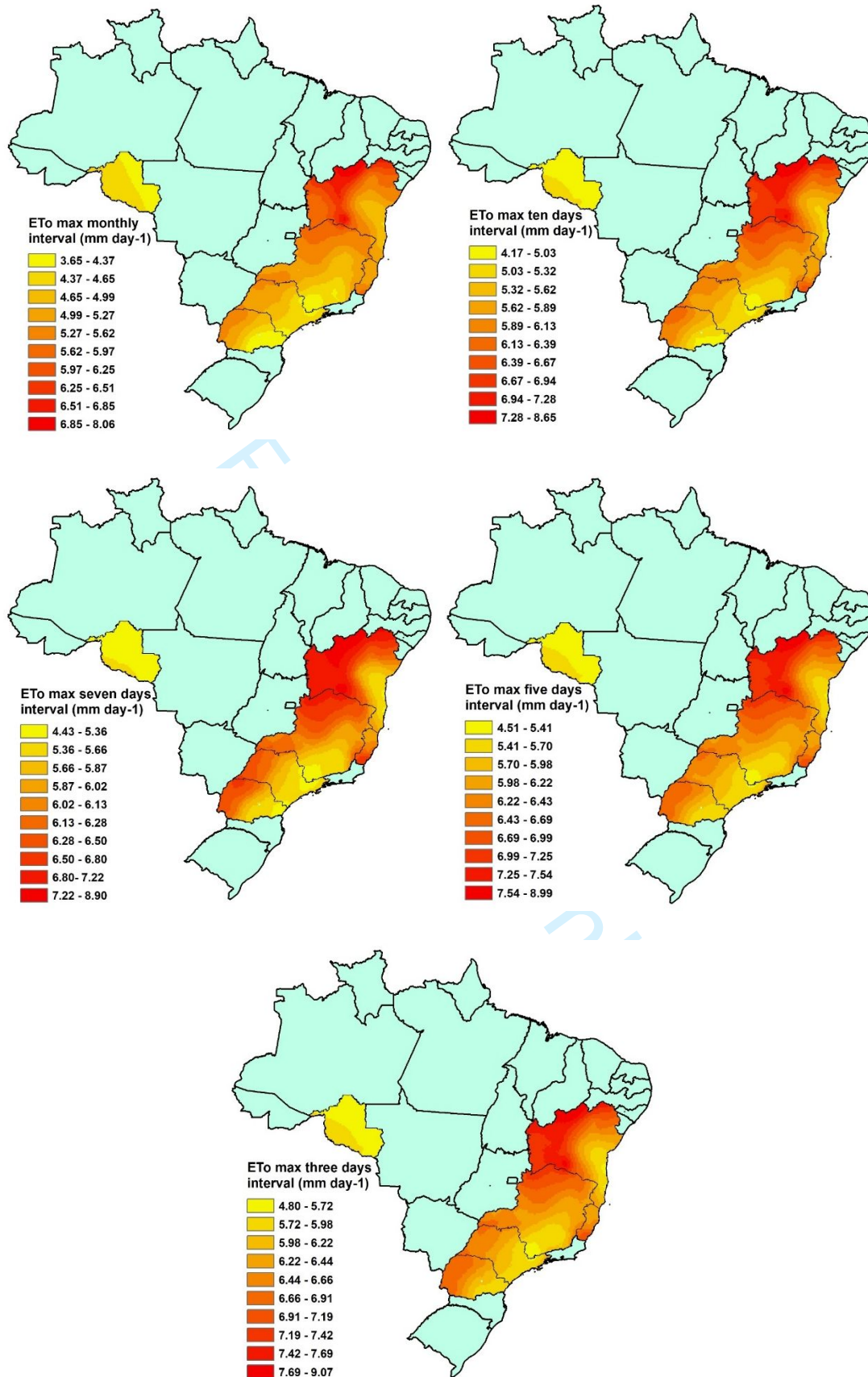


Figure 3: Spatial representation of the maximum daily reference evapotranspiration of the state of Minas Gerais, for the monthly interval and three days

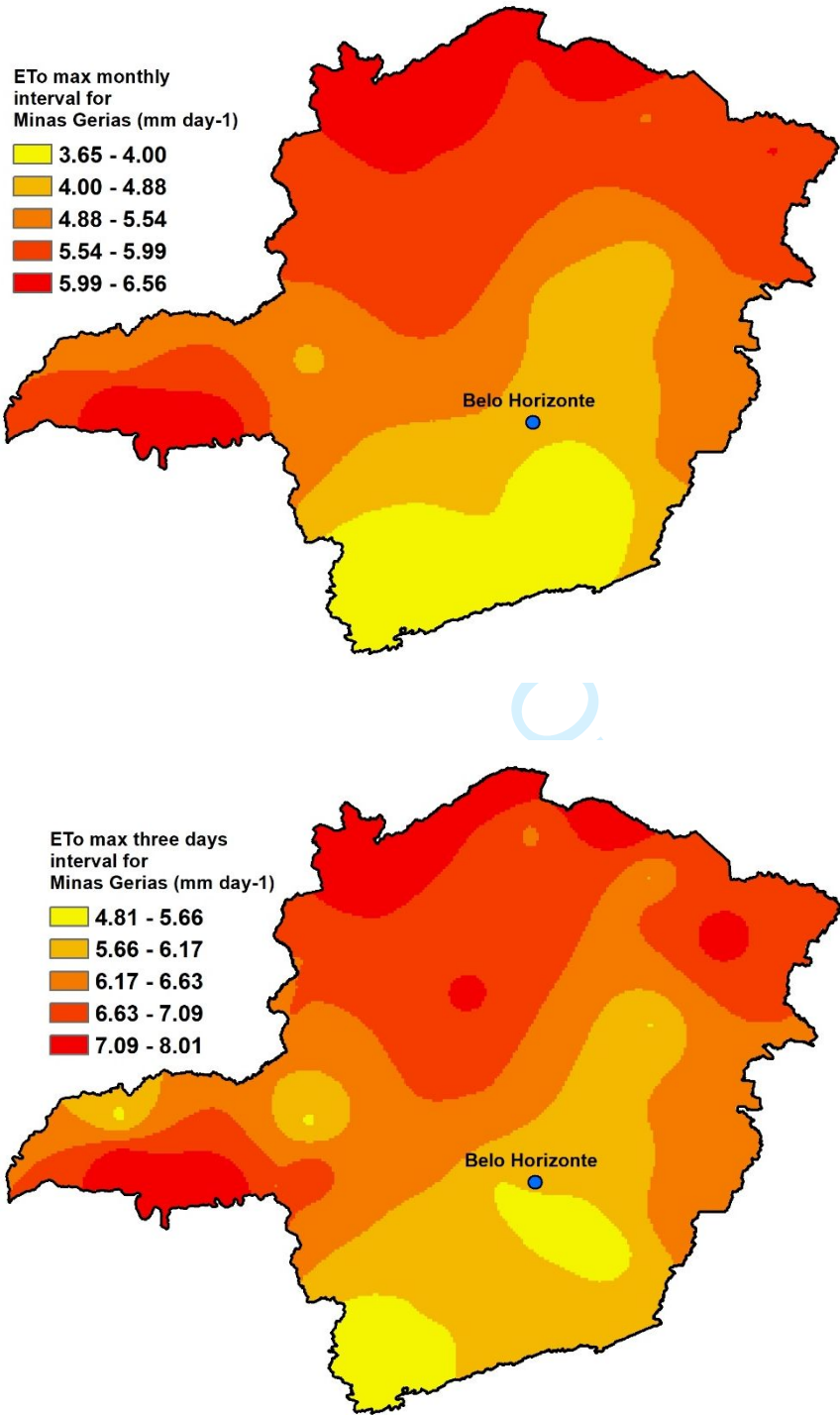


Figure 4: Spatial representation of adjustment factor to correct monthly ET to 7 days moving average.

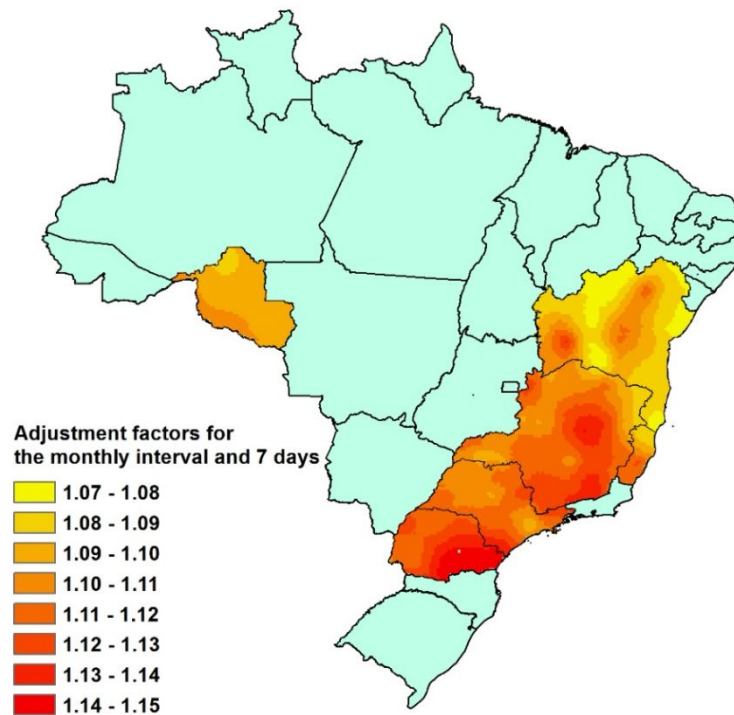


Figure 5: Spatial representation of adjustment factor to correct monthly ET to 3 days moving average.

